Evaluation Traps and Failure Analysis for RAG

Designing tests that reveal weaknesses instead of hiding them

Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

1. The Problem with Easy Tests

A system can look excellent when every question copies a sentence from the source document. Real users ask paraphrased, incomplete, comparative, and sometimes unanswerable questions. An evaluation set should challenge the retrieval and generation pipeline without becoming arbitrary. The goal is to discover failure modes early and make future comparisons reliable.

2. Categories of Test Questions

Use direct factual questions, paraphrases, questions about relationships between concepts, multi-part questions, questions requiring evidence from more than one passage, and unsupported questions. Include terms that appear only once and details near chunk boundaries. These categories test different capabilities and can reveal whether a system is relying on superficial keyword overlap.

3. Unsupported or “Trap” Questions

A trap question asks for information that the indexed corpus does not establish. For example, a collection about transformer architecture cannot support a precise claim about a particular company’s private deployment unless that fact appears in the documents. The expected behavior should be an explicit statement that the available sources do not answer the question. Do not treat a plausible guess as a correct response.

4. Labeling Expected Evidence

For each question, record the expected answer or abstention behavior and the source passages that support it. When chunk IDs change after re-ingestion, relabel or map the evidence carefully instead of assuming that old IDs remain valid. Keep the source passage and rationale with each label so another person can understand why the answer is expected.

5. Diagnose the Stage That Failed

If no relevant text appears in extracted output, the issue is extraction. If the relevant content exists but is fragmented, inspect chunking. If the right chunk exists but is not returned, inspect embedding compatibility, query wording, similarity metric, filters, and top-k. If relevant evidence is retrieved but the response invents details, inspect prompt construction, model behavior, and output validation. This stage-based approach prevents changing unrelated components blindly.

6. Reporting Results Honestly

Report the number of questions, the kinds of questions, the retrieval metric, the answer-quality criteria, and the model configuration. A result such as “100% on 18 hand-written questions” describes that specific test set; it does not prove universal accuracy. Preserve failures as regression tests. After every meaningful change, rerun the same set and review newly introduced errors as well as improvements.

Key Takeaways

Good evaluation deliberately includes difficult and unsupported questions. Label the evidence, keep the dataset stable, report its limits, and use error analysis to identify which pipeline stage needs improvement.

End of document • Use source passages as evidence and verify claims during evaluation.